

Cosmology

Exercises Sheet 1

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Exercise 1

The equation of motion of a particle moving in one dimension with total energy zero and potential $V(a)$ is

$$\ddot{a} + \frac{dV(a)}{da} = 0. \quad (1)$$

The potential $V(a)$ is as follows

$$V(a) = -\frac{4\pi G}{3} \frac{\rho_{m,0}}{a} + \frac{k}{2} - \frac{\Lambda}{6} a^2. \quad (2)$$

Therefore, the equation of motion to show from the first equation of Friedmann is

$$\ddot{a} + \left(\frac{4\pi G}{3} \frac{\rho_{m,0}}{a^2} - \frac{\Lambda}{3} a \right) = 0. \quad (3)$$

The first Friedmann equation with pressureless matter, a cosmological constant and spatial curvature is

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{c^2 \Lambda}{3}. \quad (4)$$

Differentiating the Friedmann equation (4) over time we obtain:

$$2\dot{a}\ddot{a} = \frac{8\pi G}{3}(\dot{\rho}a^2 + 2\rho a\dot{a}) + \frac{2c^2\Lambda}{3}a\dot{a} \quad (5)$$

$$\ddot{a} = \frac{4\pi G}{3} \left(\dot{\rho} \frac{a^2}{\dot{a}} + 2\rho a \right) + \frac{c^2\Lambda}{3}a \quad (6)$$

Using the conservation of energy, that is

$$\dot{\rho} + 3\frac{\dot{a}}{a} \left(\rho + \frac{p}{c^2} \right) = 0, \quad (7)$$

we can rewrite the equation (6), remembering that we have a pressureless matter, in this way:

$$\ddot{a} = \frac{4\pi G}{3}(-3\rho a + 2\rho a) + \frac{c^2\Lambda}{3}a \quad (8)$$

$$\ddot{a} = \left(-\frac{4\pi G}{3}\rho + \frac{c^2\Lambda}{3} \right) a \quad (9)$$

Here, ρ should be understood as the sum of all contributions to the energy density in the universe. We consider only the contribution by matter ρ_m , that is related to the factor scale as

$$\frac{\rho_m}{\rho_{m,0}} = \left(\frac{R}{R_0} \right)^{-3} = a^{-3}. \quad (10)$$

Substituting into the equation (9) we find the solution (with $c = 1$ in natural units):

$$\ddot{a} = -\frac{4\pi G}{3} \frac{\rho_{m,0}}{a^2} + \frac{\Lambda}{3} a. \quad (11)$$

Exercise 2

In a spatially flat model $\Omega_{m,0} + \Omega_{\Lambda,0} = 1$, the first Friedmann equation in terms of density parameters is

$$H^2 = H_0^2 \left(\frac{\Omega_{m,0}}{a^3} + \Omega_{\Lambda,0} \right), \quad (12)$$

where H is the Hubble paramter defined as $H = \dot{a}/a$. To find the age of the Universe, we solve the differential equation given by the first Friedmann equation. We start multiplying by a^2

$$\dot{a}^2 = H_0^2 \left(\frac{\Omega_{m,0}}{a} + \Omega_{\Lambda,0} a^2 \right). \quad (13)$$

Thus, the differential equation is

$$\frac{da}{H_0 \sqrt{\frac{\Omega_{m,0}}{a} + \Omega_{\Lambda,0} a^2}} = dt. \quad (14)$$

From here, we find the following integral to solve:

$$t = \int \frac{da}{H_0 \sqrt{\frac{\Omega_{m,0}}{a} + \Omega_{\Lambda,0} a^2}} \quad (15)$$

$$t = \int \frac{da}{H_0 \sqrt{\frac{\Omega_{m,0}}{a} \left(1 + \frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} a^3 \right)}} \quad (16)$$

Using the following substitution

$$u^2 = \frac{\Omega_{\Lambda,0}}{\Omega_{m,0}} a^3 \longrightarrow u = \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}} a^{\frac{3}{2}} \longrightarrow du = \frac{3}{2} \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}} a^{\frac{1}{2}} da, \quad (17)$$

we have the integral:

$$t = \frac{2}{3H_0 \sqrt{\Omega_{\Lambda,0}}} \int \frac{du}{\sqrt{1+u^2}} = \frac{2}{3H_0 \sqrt{\Omega_{\Lambda,0}}} \sinh^{-1} u \quad (18)$$

Using the relation $\sinh^{-1} u = \ln(u + \sqrt{u^2 + 1})$ [1], we find the result:

$$t = \frac{2}{3H_0\sqrt{\Omega_{\Lambda,0}}} \ln \left(\sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}} a^3 + \sqrt{\frac{\Omega_{\Lambda,0}}{\Omega_{m,0}}} a^3 + 1 \right) \quad (19)$$

Exercise 3

First of all, we show how $(x = \Omega_m(a), y = \Omega_\Lambda(a))$ satisfy the homogeneous system of equations:

$$\begin{cases} x' = -x(1 - x + 2y) \\ y' = y(x + 2 - 2y) \end{cases} \quad (20)$$

where $()' = \frac{d}{d \log(a)}$. We start with the definition of evolution of density parameters:

$$\Omega_i(t) = \frac{8\pi G}{3H^2(t)} \rho_i(t) \quad (21)$$

where the label i denotes $i = m, \Lambda$. We should ignore any contribution coming from radiation, i.e. set $\Omega_r = 0$. It's clear that each density is, in general, a function of cosmic time t . Thus, if we are interest to investigate the evolution of these densities as the universe expand, it's a good idea to derive over time:

$$\dot{\Omega}_i = \frac{8\pi G}{3H^2} \left(\dot{\rho}_i - 2 \frac{\dot{H}}{H} \rho_i \right). \quad (22)$$

Now we recall the energy conservation for barotropic fluid ($p = w\rho c^2$):

$$\dot{\rho}_i + 3H(\rho_i + w_i \rho_i) = 0 \quad (23)$$

where we have written $H = \dot{a}/a$. Thus, (22) becomes:

$$\dot{\Omega}_i = -\Omega_i H \left[3(1 + w_i) + 2 \frac{\dot{H}}{H^2} \right] \quad (24)$$

We now need an expansion for $\dot{H}$ in term of Ω_i :

$$\dot{H} = \frac{d}{dt} \left(\frac{\dot{a}}{a} \right) = \frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a} \right)^2 = \frac{\ddot{a}}{a} - H^2, \quad (25)$$

and so, using the second Friedmann equation we write:

$$\frac{\dot{H}}{H^2} = \frac{a\ddot{a}}{\dot{a}^2} - 1 = -q - 1 = -\frac{1}{2}\Omega_m + \Omega_\Lambda - 1 \quad (26)$$

where in the last passage we use the second Friedmann equation with q the deceleration parameter. Substituting this result into (24), we obtain the relation:

$$\dot{\Omega}_i = \Omega_i H (\Omega_m - 2\Omega_\Lambda - 1 - 3w_i) \quad (27)$$

Setting $w_i = 0, -1$ respectively for matter ($i = m$) and vacuum ($i = \Lambda$), we thus obtain:

$$\begin{cases} \frac{d\Omega_m}{dt} = \Omega_m H [\Omega_m - 1 - 2\Omega_\Lambda] \\ \frac{d\Omega_\Lambda}{dt} = \Omega_\Lambda H [\Omega_m - 2(\Omega_\Lambda - 1)] \end{cases} \quad (28)$$

But here the derivatives are in time, so we make H explicit and integrate over time dt :

$$\begin{cases} d\Omega_m = \Omega_m \frac{da}{a} [\Omega_m - 1 - 2\Omega_\Lambda] \\ d\Omega_\Lambda = \Omega_\Lambda \frac{da}{a} [\Omega_m - 2(\Omega_\Lambda - 1)] \end{cases} \implies \begin{cases} \frac{d\Omega_m}{da} a = \Omega_m [\Omega_m - 1 - 2\Omega_\Lambda] \\ \frac{d\Omega_\Lambda}{da} a = \Omega_\Lambda [\Omega_m - 2(\Omega_\Lambda - 1)] \end{cases} \quad (29)$$

In general $x \frac{df}{dx} = \frac{df}{d \log(x)}$ so:

$$\begin{cases} \frac{d\Omega_m}{d \log(a)} = \Omega_m [\Omega_m - 1 - 2\Omega_\Lambda] \\ \frac{d\Omega_\Lambda}{d \log(a)} = \Omega_\Lambda [\Omega_m - 2(\Omega_\Lambda - 1)] \end{cases} \implies \begin{cases} x' = x(x - 1 - 2y) \\ y' = y(x - 2y + 2) \end{cases} \quad (30)$$

Now for calculating the critical points we recall that a critical point is define as the points (x, y) such that $x' = 0$ and $y' = 0$. So, in our case, we obtain that:

$$\begin{cases} x(x - 1 - 2y) = 0 \\ y(x - 2y + 2) = 0 \end{cases} \quad (31)$$

If $x \neq 0$ we obtain:

$$\begin{cases} x = 1 + 2y \\ y(1 + 2y - 2y + 2) = 0 \end{cases} \implies \begin{cases} x = 1 + 2y \\ y = 0 \end{cases} \quad (32)$$

Thus, the critical point is $(1, 0)$. If $y \neq 0$:

$$\begin{cases} x(x - 1 - x - 2) = 0 \\ y = \frac{x+2}{2} \end{cases} \implies \begin{cases} x = 0 \\ y = \frac{x+2}{2} \end{cases} \quad (33)$$

Thus, the other critical point is $(0, 1)$. Finally, if $x = 0$ we obtain the critical point $(0, 1)$, that we just got, and the critical point $(0, 0)$. If $y = 0$ we find the points already obtained. The cosmic flow, that is given by the solutions of the homogeneous system of equations (20), is showed in the picture (1). The picture depends on the critical points. To understand the nature of the critical points we calculate the Jacobian matrix associated with the system (20). Defining $x' = f(x, y)$ and $y' = g(x, y)$:

$$J = \begin{pmatrix} \frac{\partial f}{\partial x} & \frac{\partial f}{\partial y} \\ \frac{\partial g}{\partial x} & \frac{\partial g}{\partial y} \end{pmatrix} = \begin{pmatrix} -1 + 2x - 2y & -2x \\ y & x + 2 - 4y \end{pmatrix} \quad (34)$$

Then, we evaluate the matrix at each critical point and calculate the corresponding eigenvalues. In the case of eigenvalues with the same sign the critical point is a source if $\lambda_2 > \lambda_1 > 0$ or it is a sink if $\lambda_2 < \lambda_1 < 0$. In the case of eigenvalues of opposite sign $\lambda_1 < 0 < \lambda_2$ there is a saddle point. In our case we get:

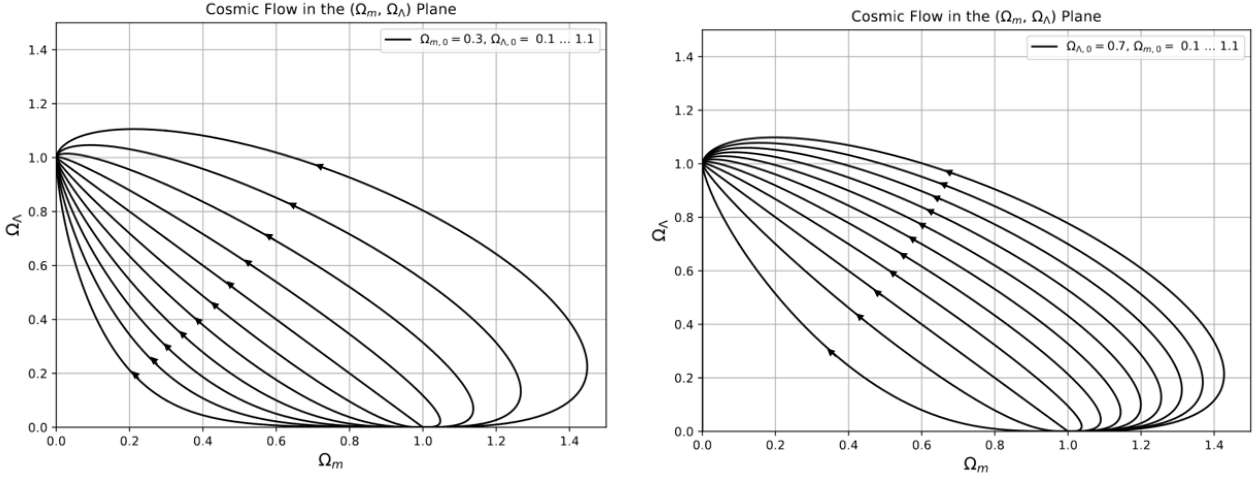


Figure 1: Evolution of density parameters Ω_m and Ω_Λ for various cosmological models passing through the points $\Omega_{m,0} = 0.3$ and $\Omega_{\Lambda,0} = 0.1 \dots 1.1$ (left-hand panel) and $\Omega_{\Lambda,0} = 0.7$ and $\Omega_{m,0} = 0.1 \dots 1.1$ (right-hand panel). The points was chosen since any general point in the plane defines a unique trajectory passing through that point, it is convenient to specify each trajectory by the present-day values $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$. The plots was done in python, using the library *scipy.integrate.odeint*.

1. Critical point $(1,0)$: eigenvalues $\lambda_1 = 1, \lambda_2 = 3$. This is a source.
2. Critical point $(0,1)$: eigenvalues $\lambda_1 = -2, \lambda_2 = -3$. This is a sink.
3. Critical point $(0,0)$: eigenvalues $\lambda_1 = -1, \lambda_2 = 2$. This is a saddle point.

Moreover, the Jacobian matrix gives the best linear approximation as x and y vary. We define the linearization of the system of equation (20) as $\dot{\mathbf{x}} = J\mathbf{x}$ and its general solution is $\mathbf{x}(t) = \alpha \mathbf{v}_1 e^{\lambda_1 t} + \beta \mathbf{v}_2 e^{\lambda_2 t}$ with α and β arbitrary coefficient and $\mathbf{v}_1$ and $\mathbf{v}_2$ eigenvectors. However this is only the analytical solution of the problem, so we move on to discuss the qualitative trend of the trajectory, in which we are interesting. Indeed, it's possible to draw the trajectory in the phase space diagram with the information given above. We know that the trajectory lead out from the critical point when the critical point is source and lead in when the critical point is sink. Therefore, as can be seen from the Figure (1), the critical points $(1,0)$ and $(0,1)$ act as 'attractors' for the trajectories. We see that the trajectories all start in $(1,0)$, which is an unstable fixed point, and converge on $(0,1)$, which is a stable fixed point. It means that the universe evolves towards the sink, that is a stable point.

Exercise 4

It's not possible that the universe has suffered a bounce since decoupling. A bounce corresponds to a Universe that has suffered first contraction and then expansion. If we assume that there really was a bounce, this would imply a maximum observable redshift. In fact, now we observe

that the Universe is expanding, but if we know that before there was a contraction it's mean that there was a moment that the Universe stopped contraction and began expansion. Therefore, there was a moment in which $\dot{R} = 0$, so the first bounce condition is $H = 0$. The second condition that we can assume is that $H' < 0$, where here we prefer to differentiate the time derivative (equipped with a dot) with the derivative in $\frac{1}{a}$ (equipped with a comma). Indeed, the bounce condition is $\dot{H} > 0$ which translates into $H' < 0$. Now we have all ingredients to show that there our Universe has not suffered a bounce. We start with the first Friedmann equation:

$$H^2(t) = H_0^2 [\Omega_{m,0}(1+z)^3 + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}] \quad (35)$$

where we ignore any contribution coming from radiation ($\Omega_{r,0} = 0$). Let's apply to this equation the first condition, assuming that exists z_{max} :

$$\Omega_{m,0}(1+z_{max})^3 + \Omega_{k,0}(1+z_{max})^2 + \Omega_{\Lambda,0} = 0 \quad (36)$$

Now we can use the cosmic sum rule $\Omega_{m,0} + \Omega_{k,0} + \Omega_{\Lambda,0} = 1$ for write (36) in term of only two parameters:

$$\Omega_{m,0}(1+z_{max})^3 + (1 - \Omega_{m,0} - \Omega_{\Lambda,0})(1+z_{max})^2 + \Omega_{\Lambda,0} = 0 \quad (37)$$

Now we need the second condition, that is $H' < 0$. Knowing that the redshift is define as $(z+1) = \frac{1}{a}$, it's possible rewrite the second condition as:

$$\left. \frac{dH}{d\left(\frac{1}{a}\right)} \right|_{t=t_s} = \left. \frac{dH}{d(1+z)} \right|_{t=t_s} = \frac{H_0}{2} \frac{3\Omega_{m,0}(1+z_{max})^2 + 2\Omega_{k,0}(1+z_{max})}{\sqrt{\Omega_{m,0}(1+z)^3 + \Omega_{k,0}(1+z)^2 + \Omega_{\Lambda,0}}} < 0 \quad (38)$$

Now it would be nice to be able to compare the two equations found (37) and (38), but to do this we must first rewrite (38) as a function of $\Omega_{m,0}$ and $\Omega_{\Lambda,0}$ (removing the denominator which is certainly positive):

$$\begin{aligned} 3\Omega_{m,0}(1+z_{max})^2 + 2(1 - \Omega_{m,0} - \Omega_{\Lambda,0})(1+z_{max}) &< 0 \\ 3\Omega_{m,0}(1+z_{max}) + 2(1 - \Omega_{m,0} - \Omega_{\Lambda,0}) &< 0 \end{aligned} \quad (39)$$

Now we can compare (37) with this last inequality. In other words, we make $\Omega_{\Lambda,0}$ explicit from the first equation:

$$\Omega_{\Lambda,0}[1 - (1+z_{max})^2] = -\Omega_{m,0}(1+z_{max})^3 - (1+z_{max})^2 + \Omega_{m,0}(1+z_{max})^2 \quad (40)$$

We can collect $(1+z_{max})^2$ and divided by $1 - (1+z_{max})^2$ and we obtain:

$$\Omega_{\Lambda,0} = \frac{-\Omega_{m,0}(1+z_{max})^3 - (1+z_{max})^2 + \Omega_{m,0}(1+z_{max})^2}{1 - (1+z_{max})^2} \quad (41)$$

We plug this equation inside (39):

$$\begin{aligned} 3\Omega_{m,0}(1+z_{max}) + 2 \left[1 - \Omega_{m,0} - \left(\frac{\Omega_{m,0}(1+z_{max})^3 + (1+z_{max})^2 - \Omega_{m,0}(1+z_{max})^2}{(1+z_{max})^2 - 1} \right) \right] &< 0 \\ 3\Omega_{m,0}(1+z_{max}) + 2 \left[\frac{-1 + \Omega_{m,0} - \Omega_{m,0}(1+z_{max})^3}{(1+z_{max})^2 - 1} \right] &< 0 \\ 3\Omega_{m,0}(1+z_{max})^3 - 3\Omega_{m,0}(1+z_{max}) - 2 + 2\Omega_{m,0} - 2\Omega_{m,0}(1+z_{max})^3 &< 0 \\ \Omega_{m,0}(1+z_{max}^3 + 3z_{max}^2 + 3z_{max} - 3 - 3z_{max} + 2) &< 2 \end{aligned}$$

At the end we obtain that:

$$\boxed{z_{max}^3 + 3z_{max}^2 < \frac{2}{\Omega_{m,0}}} \quad (42)$$

Now, if is true this equation, and knowing that $\Omega_{m,0} \approx 0.3$ implies that:

$$z_{max}^2(z_{max} + 3) < \frac{20}{3}. \quad (43)$$

For just $z_{max} = 3$ the relation is not respected. Moreover, we observe object with redshift $z > 3$ and so we can conclude that our Universe has not suffered a bounce since decoupling.

References

- [1] Eric W. Weisstein. Inverse hyperbolic functions. Wolfram Mathworld. Retrieved 2020-08-30.